

This document outlines the architecture for a custom Shopify storefront to sell **Chainmail's** white-label boxed IoT weather sensors and automate the onboarding process for new nodes joining the parametric insurance grid.

1. Product Catalog Architecture

The storefront will offer pre-packaged IoT kits and individual expansion sensors, white-labeled under the **Chainmail Sentinel** brand.

Core Bundles & Hardware Specifications

1. Sentinel Gateway Hub (Model S-100)

- *Based on Ecowitt GW1100/GW1200*
- Central receiver with built-in barometric pressure, indoor temperature, and humidity sensors.
- Connects to local Wi-Fi and pushes encrypted telemetry to the `/api/ingest/ecowitt` endpoint.
- **Price:** \$49.00 USD

1. Wittboy CAT-7 Sensor Array (Model S-700)

- *Based on Ecowitt Wittboy (GW2001)*
- Solar-powered, IP65-rated 7-in-1 outdoor sensor array.
- Measures wind speed (anemometry), wind direction, temperature, humidity, rainfall (hydrology), light, and UV index. Features ultrasonic wind sensor for zero moving parts.
- **Price:** 99.00 USD

1. Sentinel Soil Probe (Model S-WH51)

- *Based on Ecowitt WH51*
- Wireless soil moisture sensor to evaluate regional drought risks and flash flooding saturation.
- **Price:** 4.00 USD

1. Sentinel Smart Valve Actuator (Model S-WFC01)

- *Based on Ecowitt WFC01*

- Smart water timer valve designed to mitigate localized wildfire threat by auto-wetting property grounds when dry soil moisture thresholds trigger parametric alarm states.
- **Price:** \$79.00 USD

2. Automated Network Onboarding Workflow

The Shopify purchase acts as the initial transaction in the node onboarding lifecycle.

```
graph TD
    mermaid
```

```
sequenceDiagram
```

```
autonumber
```

```
actor User as Policyholder / Buyer
```

```
participant Shop as Shopify Storefront
```

```
participant Webhook as Shopify Webhook Worker
```

```
participant API as Chainmail API Server
```

```
participant DB as Firestore Database
```

```
participant Device as IoT Sentinel Device
```

```
User->>Shop: Purchase Sentinel Kit (S-100 + S-700)
```

```
Shop-->>User: Order Confirmation & Shipping Label Generated
```

```
Shop->>Webhook: Event: order/paid (Payload contains email, address)
```

```
Webhook->>API: POST /api/onboarding/provision
```

```
activate API
```

```
API->>API: Generate secure device key & pre-provision propertyId
```

```
API->>DB: Write property & pre-provisioned sensors as PENDING_ACTIVATION
```

```
API-->>Webhook: Provisioning Success
```

deactivate API

Webhook-->>User: Onboarding Email with Device Credentials & Claim QR Code

User—>>Device: Mount hardware & connect to home Wi-Fi

Device—>>API: POST /api/ingest/ecowitt (First Telemetry Ping)

activate API

API—>>API: Authenticate device with secure pre-provisioned key

API—>>DB: Update status to VERIFIED_NODE & Activate Parametric Coverage

API-->>Device: Acknowledge (HTTP 200)

deactivate API

API—>>User: App Push Notification: "Node Active. Coverage Live!"

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3. API Provisioning Endpoint Specifications

When a checkout completes, Shopify sends an `order/paid` webhook to our backend API server:

Endpoint: `POST /api/onboarding/provision`

Access Control: HMAC Signature verified using Shopify API Webhook Shared Secret.

Request Payload (from Shopify):

``json`

`{`

`"order_id": 987654321,`

`"email": "customer@example.com",`

`"shipping_address": {`

`"address1": "12665 Julington Oaks Dr",`

```
"city": "Jacksonville",

"province": "FL",

"zip": "32223",

"country": "US"

},

"line_items": [

{

"sku": "SENTINEL-KIT-A",

"quantity": 1,

"name": "Sentinel Gateway & Wittboy Array"

}

]

}
```

Provisioning Logic:

1. **Address Validation:** Call Google Address Validation API to verify CASS status and retrieve exact rooftop coordinates (lat / lng).

2. **Database Record Creation:**

- Create a new document in /properties:

```
`typescript

{

ownerEmail: "customer@example.com",

address: "12665 Julington Oaks Dr, Jacksonville, FL 32223",

lat: 30.1421936,
```

```
lng: -81.6157784,

riskScore: 72.4, // Calculated by Underwriting Agent based on coordinates

status: "PENDING_ACTIVATION",

createdAt: serverTimestamp()

}
```

- Create a document in `/properties/{propertyId}/sensors/Anemometry:`

```
`typescript

{

type: "Anemometry",

verificationStatus: "UNVERIFIED",

hardwareType: "Ecowitt",

value: 0

}
```

1. **Secret Generation:** Generate a unique password/secret token for the gateway and email it as a QR code to the user.

4. UI/UX Storefront Integration

The Shopify storefront theme will match Chainmail's core brand design system: Cobalt & Gold / Slate & Gold.

Theme Design System

- **Primary Background:** #030712 (Slate Black)
- **Secondary Cards:** rgba(17, 24, 39, 0.7) (Translucent Dark Blue)
- **Accents:** #D4AF37 (Metallic Gold)

- **Fonts:** Outfit (Headings) and Inter (Body)

Custom Liquid Elements

- **Network Node Counter:** A live ticker in the header showing: `ACTIVE_NODES: 1,482 | TOTAL_COVERAGE_LOCKED: $34,891,000``.
- **Map Twin Preview:** An interactive WebGL globe on the landing page showing glowing gold dots representing active coverage sensors in real time.
- **onboarding-tab widget:** Built-in section on the product page showing the simple "Plug, Scan, Insure" 3-step setup guide.